

CS228 Logic for Computer Science

Exercise Problem Set 2

Part 1

1. Using not more than 10 lines, prove that the following sequents are valid:

- (a) $p \wedge q \vdash q \wedge p$
- (b) $(p \wedge q) \wedge r \vdash p \wedge (q \wedge r)$
- (c) $\vdash (p \wedge q) \rightarrow q$,
- (d) $p \rightarrow (q \wedge r) \vdash (p \rightarrow q) \wedge (p \rightarrow r)$.
- (e) $q \rightarrow r \vdash (p \rightarrow q) \rightarrow (p \rightarrow r)$

2. Using *only* the rules $\wedge i$, $\wedge e1$, $\wedge e2$, $\neg i$, and $\neg e$ (no other rules), prove that the following sequents are valid:

- (a) $p \wedge q \vdash \neg \neg p \wedge q$
- (b) $\neg \neg p \wedge \neg \neg q \vdash p \wedge q$
- (c) $\neg \neg(p \wedge q) \vdash \neg \neg q \wedge \neg \neg p$

3. Prove that the following sequents are valid:

- (a) $F_1 \wedge \neg F_2 \vdash \neg(F_1 \rightarrow F_2)$
- (b) $F_1 \wedge F_2 \vdash F_1 \rightarrow F_2$
- (c) $\neg F_1 \wedge \neg F_2 \vdash F_1 \rightarrow F_2$
- (d) $\neg F_1 \wedge F_2 \vdash F_1 \rightarrow F_2$

4. Using *only* conjunction and disjunction rules (no negation or implication rules are needed), prove that the following sequents are valid.

- (a) $p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$ (Distributivity)
- (b) $(p \vee q) \vee r \vdash p \vee (q \vee r)$ (Associativity)

Part 2

5. Prove that the following is valid:

$$\vdash ((A \rightarrow D) \wedge (B \rightarrow D)) \wedge (C \rightarrow D) \rightarrow ((A \vee B \vee C) \rightarrow D)$$

6. Using only the rules $\wedge i$, $\vee i$, $\neg i$, $\neg e$, $\neg\neg e$ (no $\rightarrow$ -rules are needed), prove that the following sequents are valid.

$$(a) \neg(p \wedge q) \vdash \neg p \vee \neg q \quad (\text{De Morgan})$$

$$(b) \neg(p \vee q) \vdash \neg p \wedge \neg q \quad (\text{De Morgan})$$

7. Prove that the following are valid. In parts (b) and (c), you may cite part (a) directly as an already-established result, rather than re-deriving it.

$$(a) \vdash ((p \rightarrow q) \rightarrow p) \rightarrow p \quad (\text{Peirce's law})$$

$$(b) \vdash (((p \rightarrow q) \rightarrow p) \rightarrow p) \rightarrow q \rightarrow q$$

$$(c) \vdash ((p \rightarrow q) \rightarrow q) \rightarrow ((q \rightarrow p) \rightarrow p)$$

8. Prove that the following is valid:

$$\vdash (p \rightarrow q) \vee (q \rightarrow p)$$

Give two proofs: (a) a short proof that uses LEM, and (b) a longer proof that does not cite LEM (you may still use $\neg\neg e$).

9. In natural deduction that we saw in class, we did not have separate rules for $\leftrightarrow$. Suppose we add the following $\leftrightarrow$ introduction and elimination rules:

$$\frac{\varphi \rightarrow \psi \quad \psi \rightarrow \varphi}{\varphi \leftrightarrow \psi} \leftrightarrow i \qquad \frac{\varphi \leftrightarrow \psi}{\varphi \rightarrow \psi} \leftrightarrow e_1 \qquad \frac{\varphi \leftrightarrow \psi}{\psi \rightarrow \varphi} \leftrightarrow e_2$$

where φ and ψ are wffs. Now, using natural deduction along with these extra rules, prove that the following are valid.

$$(a) \neg y \vdash x \leftrightarrow ((\neg x \wedge y) \vee (x \wedge \neg y))$$

$$(b) \vdash \neg(A \leftrightarrow \neg A)$$

$$(c) A \leftrightarrow B \vdash \neg A \leftrightarrow \neg B$$

10. Given the following statements, can you prove that the unicorn is mythical? How about magical? How about horned?

If the unicorn is mythical, then it is immortal. If it is not mythical, then it is a mortal mammal. If the unicorn is either immortal or a mammal, then it is horned. The unicorn is magical if it is horned.